IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 26 July 2025, accepted 10 September 2025, date of publication 12 September 2025,
date of current version 18 September 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3609472

RESEARCH ARTICLE

QARR-QGF: A Dual-Module Data Augmentation Framework for Enhanced Few-Shot Question Answering

SIAO WAH TAN[1], CHIN POO LEE[2], (Senior Member, IEEE), KIAN MING LIM[2], (Senior Member, IEEE), AND ALI ALQAHTANI[3,4]

[1]Faculty of Information Science and Technology, Multimedia University, Melaka 75450, Malaysia
[2]School of Computer Science, University of Nottingham Ningbo China, Yinzhou, Ningbo, Zhejiang 315100, China
[3]Department of Computer Science, King Khalid University, Abha 61421, Saudi Arabia
[4]Center for Artificial Intelligence (CAI), King Khalid University, Abha 61421, Saudi Arabia

Corresponding authors: Chin Poo Lee (Chin-Poo.Lee@nottingham.edu.cn) and Kian Ming Lim (Kian-Ming.Lim@nottingham.edu.cn)

This work was supported by the Deanship of Scientific Research at King Khalid University, Saudi Arabia, under Grant RGP2/314/45.

ABSTRACT Few-shot question answering (QA) aims to train effective QA models using very limited annotated data, but existing approaches often struggle to generalize due to insufficient training examples. This paper proposes a two-stage data augmentation framework, called Question-Answer Replacement and Removal and Question Generation and Filtering (QARR-QGF), to address this challenge. The QARR module enhances pretraining data by systematically replacing and removing question-answer pairs to create more diverse examples. During fine-tuning, the QGF module generates paraphrased questions and applies semantic filtering to retain high-quality training samples. The framework is evaluated using three widely used generative models: Longformer-Encoder-Decoder (LED), BART, and T5, on the SQuAD, HotpotQA, and Natural Questions datasets. Experimental results show that QARR-QGF consistently improves performance across all datasets and few-shot settings. For example, the QARR-QGF-T5 model achieves F1 scores of 82.3% on SQuAD, 59.9% on HotpotQA, and 59.0% on Natural Questions in the 16-shot setting, outperforming previous state-of-the-art methods. These results demonstrate the effectiveness of QARR-QGF in improving few-shot QA performance by generating richer and more diverse training data.

INDEX TERMS Few-shot, question answering, deep learning, language model, generative model.

I. INTRODUCTION

Few-shot question answering refers to the task of training language models to generate accurate and contextually appropriate answers to questions while using only a limited number of labeled training examples. This task is of significant importance in natural language processing, as it addresses the challenge of building robust systems in scenarios where labeled data is scarce. Applications of few-shot question answering span various domains, including virtual assistants, healthcare, customer support, and educational technologies, where models must respond intelligently to user queries despite limited domain-specific training data. The ability to generalize from a small dataset is critical for deploying systems in specialized or resource-constrained environments.

Recent advancements in language models have achieved remarkable progress in few-shot question answering by leveraging pretraining on large-scale datasets followed by fine-tuning on small, task-specific datasets. However, this approach faces inherent challenges. One major limitation is overfitting to the small fine-tuning dataset, which restricts the model's generalization capability. Additionally, the lack of diversity in training samples often results in models that are overly sensitive to linguistic variations, leading to reduced performance on unseen data. Moreover, the mismatch between the pretraining and fine-tuning phases further exacerbates this issue, as models pretrained on general-purpose datasets struggle to adapt effectively to the requirements of specific tasks.

The associate editor coordinating the review of this manuscript and approving it for publication was Arianna DUlizia.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

The motivation for the proposed method stems from these challenges. Addressing the limitations of existing approaches requires a robust mechanism for augmenting training data while preserving semantic integrity. A well-designed augmentation strategy can mitigate overfitting, enhance generalization, and bridge the gap between pretraining and fine-tuning phases. To this end, this study introduces the Question-Answer Replacement and Removal and Question Generation and Filtering (QARR-QGF) framework. This dual-module framework focuses on systematic data augmentation during both the pretraining and fine-tuning phases, aiming to address the critical issues of insufficient data diversity and overfitting.

The proposed framework comprises two key modules. The Question-Answer Replacement and Removal (QARR) module operates in the pretraining phase, augmenting the dataset by generating diverse and semantically consistent question-answer pairs. This is achieved through systematic replacement and removal techniques, which enhance the model's exposure to varied linguistic patterns. In the fine-tuning phase, the Question Generation and Filtering (QGF) module generates high-quality paraphrased questions and applies semantic filtering to ensure relevance and accuracy, thus augmenting the few-shot dataset with enriched training samples. The complementary roles of these modules enable the model to achieve a deeper understanding of question-answer relationships and adapt effectively to low-resource scenarios.

This study evaluates the proposed framework using three widely recognized datasets: the Stanford Question Answering Dataset (SQuAD), HotpotQA, and Natural Questions. The results demonstrate that the framework significantly outperforms state-of-the-art methods across various few-shot settings, achieving higher F1 scores and demonstrating improved robustness. The key contributions of this paper are summarized as follows:

- A dual-module data augmentation framework, termed the Question-Answer Replacement and Removal and Question Generation and Filtering (QARR-QGF) framework, is proposed to address the challenges of few-shot question answering by systematically enhancing training data quality and diversity.
- The Question-Answer Replacement and Removal (QARR) module is designed to systematically modify the pretraining dataset by replacing and removing question-answer pairs, thereby generating a diverse set of training examples. This augmentation approach ensures robust exposure to varied linguistic and semantic structures, enhancing the model's ability to generalize.
- The Question Generation and Filtering (QGF) module is developed to address data scarcity during fine-tuning by generating semantically consistent paraphrased questions using a sequence-to-sequence architecture. A rigorous filtering mechanism is applied to retain high-quality paraphrases, ensuring alignment with the original semantic intent and improving the robustness of the fine-tuning dataset.

The remainder of this paper is structured as follows: Section II reviews the current state of research in few-shot question answering, focusing on data augmentation and learning strategies. Section III provides a detailed description of the proposed framework, including its architecture and components. Section IV describes the datasets and experimental setups, while Section V outlines the pretraining and fine-tuning processes. Section VI presents experimental results, including hyperparameter tuning, ablation study, quantitative evaluation, qualitative evaluation, and a comparison with existing methods. Finally, Section VII concludes with insights and potential directions for future research.

II. RELATED WORKS

This section reviews recent advancements in few-shot question answering, focusing on methods that address core challenges such as limited labeled data, generalization, alignment between pretraining and fine-tuning, and data augmentation. These challenges hinder the performance of large language models (LLMs) in few-shot settings, and various frameworks have emerged to mitigate these issues.

One prominent approach to tackling these challenges is the alignment of pre-trained models with fine-tuning strategies, ensuring effective knowledge transfer from pretraining to the few-shot setting. Chada and Natarajan [1] proposed FewShotQA, a framework that aligns fine-tuning with pretraining objectives to prevent performance decline. The framework uses input and output templates: the input concatenates the question, mask token, and context, while the output combines the question and answer. The method, based on the BART model, was evaluated on the MRQA shared task, achieving F1 scores of 55.5% on SQuAD, 50.5% on TriviaQA, 46.7% on Natural Questions, 38.9% on NewsQA, 39.8% on SearchQA, 45.1% on HotpotQA, 49.4% on BioASQ, and 19.9% on TextbookQA in a 16-shot setting.

Xu et al. [2] proposed the Pre-trained Machine Reader (PMR) to address the mismatch between model pretraining and downstream fine-tuning. PMR adapts pretrained masked language models (MLMs) for machine reading comprehension (MRC) tasks without additional labeled data. The architecture includes an extractor and an MLM encoder. The extractor predicts the probability score between the hidden states of two tokens produced by the encoder. PMR is pretrained on 18 million MRC-style training samples derived from Wikipedia anchors. Using RoBERTa as the encoder, PMR was evaluated on the MRQA benchmark. In a 16-shot setting, PMR achieved F1 scores of 46.5% on SQuAD, 47.7% on TriviaQA, 32.6% on NQ, 26.2% on NewsQA, 50.1% on SearchQA, 32.9% on HotpotQA, 49.1% on BioASQ, and 27.9% on TextbookQA.

Li et al. [3] introduced MRC-PASCL, a method combining noun-entity-aware post-training and answer span-oriented contrastive learning to improve low-resource machine

reading comprehension. The post-training strategy generates pseudo-MRC data by masking nouns and named entities, aligning with extractive MRC tasks. During fine-tuning, contrastive learning selects spans around the correct answer as negative examples to enhance span localization accuracy. Using the Splinter model as its backbone, MRC-PASCL achieved an F1 score of 31.60% on the TBQA dataset in a 16-shot setting.

Another contribution is from Xiao et al. [4] introduced TPKE-QA to enhance few-shot extractive question answering (EQA) by bridging the gap between pre-training and fine-tuning through post-training and knowledge-enhanced fine-tuning. The model uses the pre-trained Splinter [5] as its backbone. During post-training, EQA-style samples are generated using an entity-noun-oriented span selection strategy from unlabeled data. Knowledge-enhanced sequence generation addresses the embedding space alignment issue. TPKE-QA was evaluated on the MRQA 2019 benchmark, achieving F1 scores of 65.92% on SQuAD, 51.03% on TriviaQA, 40.31% on NQ, 40.53% on NewsQA, 52.75% on SearchQA, 49.25% on HotpotQA, 60.02% on BioASQ, and 46.85% on TBQA in a 16-shot setting.

Building on these alignment approaches, some researchers focus on enhancing the model's generalization capabilities by generating additional training data. Chen et al. [6] introduced the Generative Prompt-Based Data Augmentation (GOTTA) framework, inspired by human reasoning, to enhance few-shot question answering. GOTTA uses a public knowledge base to generate additional training samples with cloze-style questions. The process involves masking tokens in the context and combining them with a predefined template to create prompt data. The framework, based on the BART model, was tested on the MRQA shared task, achieving F1 scores of 57.8% on SQuAD, 40.8% on TriviaQA, 47.1% on Natural Questions, 36.2% on NewsQA, 41.8% on SearchQA, 45.9% on HotpotQA, 55.2% on BioASQ, and 20.5% on TextbookQA.

Chen et al. [7] introduced MINPROMPT, a minimal data augmentation framework based on unsupervised question generation and a graph-based algorithm. The framework comprises three components: the sentence graph construction, data selection, and question generation modules. Named entity recognition is first used to extract entities and relevant sentences from the raw text. In the sentence graph construction module, sentences are represented as nodes, and entities as edges. The data selection module identifies a minimal set of sentences that cover the entire text while minimizing computational resources. These selected sentences are then converted into question-answer pairs for training. Experiments using the MRQA 2019 shared task dataset and BART-large as the backbone model showed F1 scores of 73.6% on SQuAD, 50.9% on TriviaQA, 58.5% on NQ, 46.5% on NewsQA, 55.4% on SearchQA, 57.1% on HotpotQA, 57.2% on BioASQ, and 42.2% on TextbookQA in a 16-shot setting.

Moving beyond data augmentation, some methods aim to improve the reasoning process, enabling models to infer answers more effectively in few-shot settings. Haji et al. [8] introduced the Exploratory Inference Chain (EIC) framework to enhance large language models (LLMs) for multi-hop inference. The framework generates keywords from the question text using LLMs, which are then matched with relevant paragraphs. The LLM extracts information from both the paragraph and the question, and additional paragraphs are retrieved as needed to supplement the inference. All retrieved information is formatted and passed into the answering module. The framework was tested on two multi-hop inference datasets, using GPT-J-6B as the LLM. Experiments, which randomly sampled 1000 questions per dataset, showed accuracy scores of 0.334 on 2WikiMultihopQA and 0.213 on HotpotQA.

Zhang et al. [9] introduced Self-Convince, a framework designed to enhance the reasoning ability of LLMs. The framework includes three components: the Normal Chain-of-Thought (CoT), the Convincer, and the Answerer. The process begins with the Normal CoT generating an initial answer, which is then evaluated by the Convincer. If correct, the answer is clarified as final; if incorrect, the Convincer revises the reasoning path, which is provided to the Answerer to generate an improved answer. This iterative correction continues until a satisfactory answer is found. Using GPT-turbo-3.5 as the model, the framework achieved 62.0% accuracy on the AQuA dataset, a 6.1% improvement over the standard CoT.

Other approaches enhance reasoning by integrating external knowledge or search results. Luo et al. [10] introduced SAIL, a search-augmented instruction learning approach to improve LLM capabilities. SAIL addresses the limitations of static knowledge bases and in-house retrievers when connected with LLMs. The training corpus is created by concatenating instructions or GPT-4-generated responses with input queries, followed by retrieving the top three search results from the DuckDuckGo (DDG) search engine and the BM25 Wikipedia retriever. These search results are then formatted according to a predefined input prompt for training. The model achieved 52.0% accuracy on the OpenbookQA dataset in a zero-shot setting using DDG search results.

Yu et al. [11] introduced REFEED, a plug-and-play retrieval feedback pipeline designed to enhance LLM performance without costly fine-tuning. REFEED addresses the issue of outdated knowledge in LLMs by retrieving relevant information from large document collections based on the original query and generated outputs. This allows the model to reconsider and regenerate its outputs. Unlike previous methods that rely on costly human feedback, REFEED enables real-time feedback during inference. In a zero-shot setting, the method achieved an F1 score of 15.1% on the WoW dataset using Text-Davinci-003 (TD-003) as the backbone.

Zou et al. [12] proposed a resilient generative model to address overfitting and insufficiency in FSQA. Their approach incorporates two key components: a partial learning module and an answer choosing module. The partial learning module employs gradient-based importance estimation to identify and update only a crucial sub-network of the pre-trained language model (PLM), thereby reducing overfitting and preserving pretrained knowledge. The answer choosing module introduces a selective focus copy mechanism that dynamically selects the most important attention heads to copy relevant input tokens during decoding, effectively mitigating hallucinations. The model aligns the input structure with pretraining objectives by prefixing semantic roles, and optimizes training through a partial update strategy. Evaluated on eight QA benchmarks from the MRQA shared task under various few-shot settings, the proposed model consistently outperformed state-of-the-art baselines such as Gotta and FewshotQA, achieving F1 scores of 75.2% on SQuAD, 63.6% on TriviaQA, 59.2% on NQ, 50.8% on NewsQA, 57.1% on SearchQA, 62.5% on HotpotQA, 69.6% on BioASQ, and 42.7% on TextbookQA in the 16-shot setting.

Table 1 summarizes the related work in few-shot question answering. While prior methods have made progress in few-shot question answering, they exhibit notable limitations in addressing key challenges. For example, FewShotQA [1] and PMR [2] focus on aligning pretraining objectives with downstream tasks but do not adequately address overfitting or data diversity. GOTTA [6] and MINPROMPT [7] attempt to generate synthetic data, yet lack explicit mechanisms to ensure semantic consistency or control the quality of augmented samples. TPKE-QA [4] incorporates knowledge enhancement, but does not target paraphrastic variation or lexical robustness. In contrast, our proposed QARR-QGF framework introduces a dual-module augmentation strategy that systematically increases data diversity during both pretraining and fine-tuning. The QARR module enhances generalization through structured replacement and removal operations, while the QGF module filters high-quality paraphrases to maintain semantic integrity.

TABLE 1. Summary of related work in 16-shot question answering. [Acc: Accuracy, F1: F1 Score].

Method	Dataset	F1 / Acc (%)
FewShotQA [1]	SQuAD	55.5 (F1)
	TriviaQA	50.5 (F1)
	NQ	46.7 (F1)
	NewsQA	38.9 (F1)
	SearchQA	39.8 (F1)
	HotpotQA	45.1 (F1)
	BioASQ	49.4 (F1)
	TextbookQA	19.9 (F1)
Pre-trained Machine Reader (PMR) [2]	SQuAD	46.5 (F1)
	TriviaQA	47.7 (F1)
	NQ	32.6 (F1)
	NewsQA	26.2 (F1)
	SearchQA	50.1 (F1)
	HotpotQA	32.9 (F1)
	BioASQ	49.1 (F1)
	TextbookQA	27.9 (F1)
MRC-PASCL [3]	TBQA	31.6 (F1)
TPKE-QA [4]	SQuAD	65.92 (F1)
	TriviaQA	51.03 (F1)
	NQ	40.31 (F1)
	NewsQA	40.53 (F1)
	SearchQA	52.75 (F1)
	HotpotQA	49.25 (F1)
	BioASQ	60.02 (F1)
	TextbookQA	46.85 (F1)
GOTTA [6]	SQuAD	57.8 (F1)
	TriviaQA	40.8 (F1)
	NQ	47.1 (F1)
	NewsQA	36.2 (F1)
	SearchQA	41.8 (F1)
	HotpotQA	45.9 (F1)
	BioASQ	55.2 (F1)
	TextbookQA	20.5 (F1)
MINPROMPT [7]	SQuAD	73.6 (F1)
	TriviaQA	50.9 (F1)
	NQ	58.5 (F1)
	NewsQA	46.5 (F1)
	SearchQA	55.4 (F1)
	HotpotQA	57.1 (F1)
	BioASQ	57.2 (F1)
	TextbookQA	42.2 (F1)
Self-Convince Framework [9]	AQuA	62.0 (Acc)
EIC [8]	2WikiMultihopQA	33.4 (Acc)
	HotpotQA	21.3 (Acc)
SAIL [10]	OpenbookQA	52.0 (Acc)
REFEED [11]	WoW	15.1 (F1)
Resilient Model [12]	SQuAD	75.2 (F1)
	TriviaQA	63.6 (F1)
	NQ	59.2 (F1)
	NewsQA	50.8 (F1)
	SearchQA	57.1 (F1)
	HotpotQA	62.5 (F1)
	BioASQ	69.6 (F1)
	TextbookQA	42.7 (F1)

III. QARR-QGF FRAMEWORK

Few-shot question answering (FSQA) presents significant challenges, primarily due to the limited availability of training examples, which hampers a language model's ability to generalize effectively. Language models typically undergo a pretraining phase on large-scale datasets, followed by fine-tuning on smaller few-shot datasets. However, the scarcity of examples in the few-shot phase exacerbates the difficulty of generalization, leading to reduced performance in practical scenarios.

To address these challenges posed by the constrained sample size in FSQA and to enhance the generalization capabilities of language models, we propose the QARR-QGF framework, a dual-module data augmentation approach. The framework consists of two key components: (1) the Question-Answer Replacement and Removal (QARR) module, which augments the dataset during the pretraining phase, and (2) the Question Generation and Filtering (QGF) module, which augments the dataset during the fine-tuning phase. Both modules work synergistically to improve the model's ability to generalize from limited data, contributing to better performance in FSQA tasks.

The QARR module enriches the pretraining data by systematically replacing or removing question-answer pairs, thus creating a more diverse and robust dataset for the initial training phase. On the other hand, the QGF module, applied

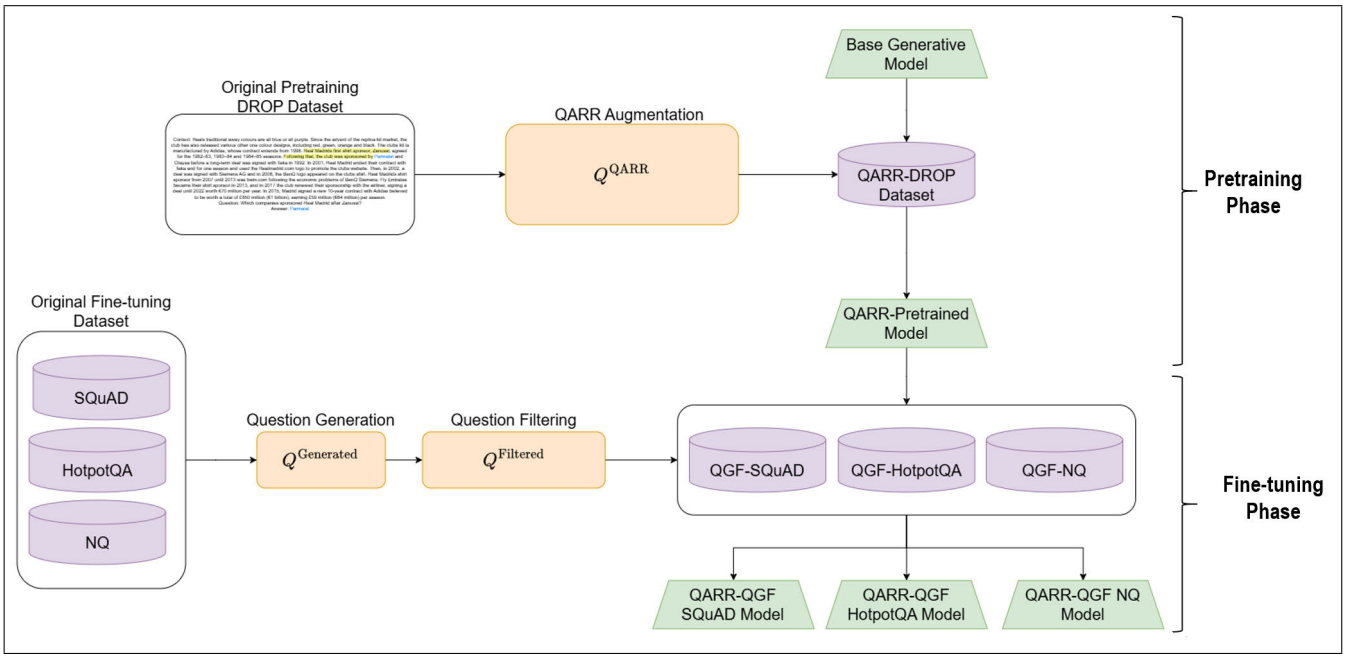


FIGURE 1. Overview of the QARR-QGF framework for few-shot question answering. In the pretraining phase, the QARR module augments the DROP dataset through question-answer replacement and word removal, producing the QARR-DROP dataset. In the fine-tuning phase, the QGF module applies question generation and filtering to augment existing few-shot training datasets (SQuAD, HotpotQA, and NQ), yielding the final augmented datasets QGF-SQuAD, QGF-HotpotQA, and QGF-NQ for downstream few-shot training.

during the fine-tuning phase, generates paraphrased questions and applies a filtering mechanism to ensure the quality and relevance of the augmented samples. This process helps mitigate the underfitting issues often encountered in few-shot learning scenarios, enhancing the overall performance of the model. Figure 1 illustrates the overall process flow of the QARR-QGF framework.

A. QUESTION-ANSWER REPLACEMENT AND REMOVAL (QARR) FOR AUGMENTATION IN PRETRAINING

In the pretraining phase of the QARR-QGF framework, the primary goal is to establish a strong foundational understanding of question-answer pair structures and enable the model to effectively infer answers from contextual information. In this phase, question-answer pairs are organized such that the input sequence consists of the question, a mask token representing the missing answer, and the relevant context. The target sequence includes the question and the correct answer, facilitating the model's learning of the relationship between the masked input and the expected output. The input and target sequences are formally represented as:

$$\mathbf{x} = [\text{Question: } \mathbf{q}?, \text{Answer: } \mathcal{M}, \text{Context: } \mathbf{c}]$$
$$\mathbf{y} = [\text{Question: } \mathbf{q}?, \text{Answer: } \mathbf{a}]$$

where $\mathbf{q}$ denotes the original question, $\mathcal{M}$ is the mask token representing the missing answer, $\mathbf{c}$ is the context, and $\mathbf{a}$ is the correct answer.

The QARR technique enhances the pretraining dataset by generating new question-answer pairs from the existing ones. This augmentation is particularly important in few-shot scenarios, where labeled data is scarce. The QARR module consists of two stages: Question-Answer Replacement and Word Removal.

1) QUESTION-ANSWER REPLACEMENT

In the Question-Answer Replacement stage, the original question is modified by substituting specific question words with the corresponding answer. This modification generates a new question that incorporates the answer, thus improving the clarity and specificity of the question. For example, if the original question is "What is the capital of France?" and the answer is "Paris", the modified question could be transformed into "Paris is the capital of France?". The operation can be expressed as:

$$\mathbf{q}' = \text{replace}(\mathbf{q}, \text{QuestionWords}, \mathbf{a}),$$

where $\mathbf{q}$ represents the original question, $\mathbf{a}$ denotes the corresponding answer, and the function replace($\mathbf{q}$, QuestionWords, $\mathbf{a}$) substitutes the relevant question words in $\mathbf{q}$ with the answer $\mathbf{a}$.

To ensure the grammatical integrity of the modified question, the substitution process is governed by rule-based constraints that preserve syntactic compatibility. Specifically, replacements are applied only to tokens in the original question that align with named entities, noun phrases, or numeric expressions, as identified through part-of-speech (POS) tagging. This filtering mechanism helps to avoid ungrammatical constructions and ensures that the augmented questions remain both syntactically valid and contextually meaningful. Furthermore, any question-answer pairs from the DROP dataset that fail to meet these criteria are excluded

from augmentation, thereby maintaining high-quality data generation.

2) WORD REMOVAL

In the Word Removal stage, a portion of the words in the modified question is selectively removed, creating a new question–answer pair. The removed portion, denoted as $\mathbf{a}'$, becomes the new answer, and the remaining text forms the updated question. For instance, continuing with the previous example, if the modified question is "Paris is the capital of France?", removing the word "France" would result in "Paris is the capital of ?", where $\mathbf{a}' =$ "France". This process can be mathematically formulated as:

$$\mathbf{q}'' = \text{remove}(\mathbf{q}', p),$$
$$\mathbf{a}' = \text{extracted}(\mathbf{q}', p),$$

where p represents the proportion of words to be removed from the modified question $\mathbf{q}'$, and $\mathbf{a}'$ is the removed text segment treated as the new answer.

The newly generated question $\mathbf{q}''$, together with $\mathbf{a}'$, forms a new question–answer pair. The input $\mathbf{x}_{\text{QARR}}$ and target $\mathbf{y}_{\text{QARR}}$ sequences for the QARR module are then defined as:

$$\mathbf{x}_{\text{QARR}} = [\text{Question: } \mathbf{q}''?, \text{Answer: } \mathcal{M}, \text{Context: } \mathbf{c}]$$
$$\mathbf{y}_{\text{QARR}} = [\text{Question: } \mathbf{q}''?, \text{Answer: } \mathbf{a}']$$

The process of augmenting pretraining data through the QARR module is outlined in Algorithm 1, which describes the systematic steps for generating diverse question-answer pairs by replacing specific question components with answers and selectively removing words to create new pairs.

Algorithm 1 Question-Answer Replacement and Removal (QARR)

Input: Dataset $\mathcal{D} = \{(\mathbf{q}, \mathbf{a}, \mathbf{c})\}$; removal proportion p
Output: Augmented dataset $\mathcal{D}_{\text{QARR}}$

```
Initialize D_QARR ← ∅
for each (q, a, c) ∈ D do
    Replace question words in q with a to obtain q'
    Remove proportion p of words from q' to obtain q''
    Extract removed words as new answer a'
    Add (q'', a', c) to D_QARR
end for
return D_QARR
```

Figure 2 presents an example of a question-answer pair after applying the QARR module. By employing the QARR module to augment the pretraining dataset, models are exposed to a broader range of question-answer pairs, improving their ability to generalize from limited examples. This augmentation is particularly valuable in FSQA tasks, where the scarcity of training data makes generalizing to unseen questions crucial. The QARR module thus equips models with enhanced robustness, resulting in improved performance in low-resource settings.

B. QUESTION GENERATION AND FILTERING (QGF) FOR AUGMENTATION IN FINE-TUNING

Fine-tuning is critical for adapting pre-trained models to new tasks and datasets. In few-shot question answering tasks, fine-tuning is often constrained by the limited availability of training samples, typically with $k = \{16, 32, 64, 128\}$ examples per task. To address this limitation, the QGF module is introduced to augment the fine-tuning dataset with additional high-quality samples. The QGF module consists of two stages: (1) Question Generation and (2) Question Filtering.

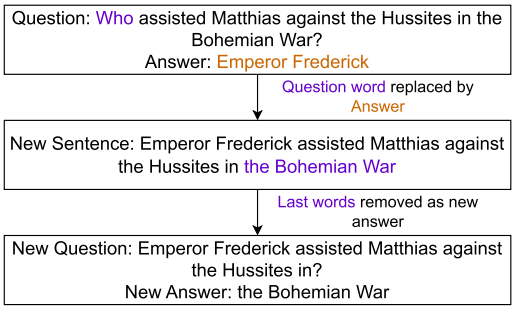


FIGURE 2. Workflow of the QARR module demonstrating question-answer replacement and word removal. The question word is first replaced by the answer to create a modified sentence. Subsequently, part of the modified sentence is truncated, with the truncated segment reassigned as the new answer to construct an alternative question-answer pair.

1) QUESTION GENERATION

The Question Generation stage utilizes the Bidirectional and Auto-Regressive Transformers (BART)[1] [13], based on `facebook/bart-large` and fine-tuned for paraphrasing to generate paraphrased questions based on existing training samples. Rather than fine-tuning BART on a specific dataset, we adopt a prompt-based generation strategy that leverages the pretrained capabilities of the model. Each input is structured in the format "`paraphrase: question sentence`", explicitly instructing the model to produce a semantically equivalent paraphrase of the given question.

BART's encoder-decoder architecture enables it to generate fluent and contextually relevant paraphrases by capturing both global and local dependencies in the input. In the encoding phase, the original question is tokenized into a sequence of tokens, which are then transformed into contextual embeddings via multiple layers of bidirectional self-attention. These embeddings are passed to the decoder, which generates the output sequence autoregressively. At each decoding step, the model predicts the next token conditioned on the previously generated tokens and the encoder output.

This paraphrasing mechanism introduces lexical and syntactic diversity into the training set, thereby improving the generalizability of few-shot question answering models. The question generation process can be mathematically expressed as:

$$t_{i+1} = \arg\max p(t_{i+1} | t_1, \ldots, t_i, \text{Context}),$$

[1] https://huggingface.co/eugenesiow/bart-paraphrase

where t_{i+1} is the next predicted token, and the output sequence is generated iteratively until an end-of-sequence token is reached. The generated paraphrased question is then passed to the Question Filtering stage for validation before being added to the augmented training set.

2) QUESTION FILTERING

In the Question Filtering stage, a filtering mechanism ensures that the generated questions are valid, relevant, and semantically similar to the original questions. This step is essential for maintaining the quality of the augmented training data. Both the original and generated questions are encoded into vector embeddings using the BART model. The cosine similarity between their embeddings is calculated and compared to a threshold τ. If the similarity exceeds the threshold, the generated question is accepted; otherwise, it is discarded. The cosine similarity is defined as:

$$\text{CosineSimilarity}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

where $\mathbf{u}$ and $\mathbf{v}$ are the embeddings of the original and generated questions, respectively. The detailed process of the QGF module is presented in Algorithm 2, which outlines the steps for generating and filtering paraphrased questions to augment the fine-tuning dataset.

Algorithm 2 Question Generation and Filtering (QGF)

Input: Dataset $\mathcal{D} = \{(\mathbf{q}, \mathbf{a}, \mathbf{c})\}$; similarity threshold τ
Output: Augmented dataset $\mathcal{D}_{\text{QGF}}$

```
Initialize D_QGF ← D
for each (q, a, c) ∈ D do
    Generate paraphrase q_gen ← BART(q)
    Compute embeddings u ← Embed(q), v ← Embed(q_gen)
    Compute similarity s ← cos(u, v)
    if s ≥ τ then
        Add (q_gen, a, c) to D_QGF
    end if
end for
return D_QGF
```

Figure 3 illustrates the QGF workflow, showcasing an example of a question being processed through the Question Generation stage using BART and subsequently evaluated in the Question Filtering stage. This diagram highlights the rigorous filtering mechanism that ensures high-quality augmented questions are added to the fine-tuning dataset, maintaining both diversity and relevance.

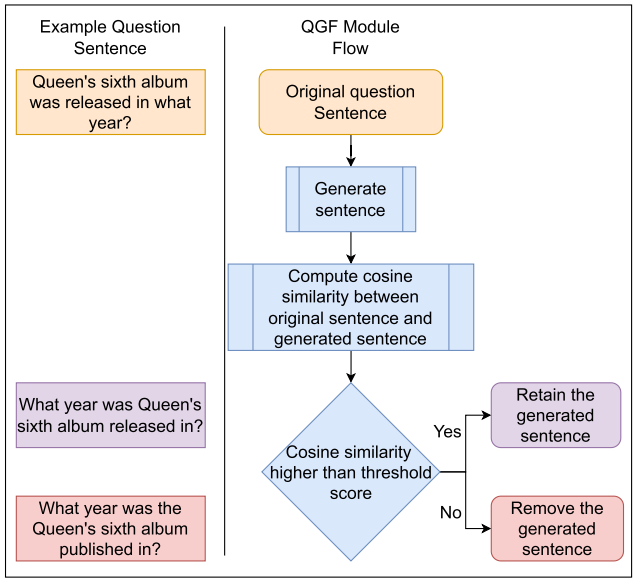


FIGURE 3. Workflow of the QGF module demonstrating question generation and filtering. An example input question is paraphrased into candidate sentences. Cosine similarity is computed between the original and generated questions, and candidates are retained or removed based on whether the similarity exceeds the dataset-specific threshold τ.

IV. QUESTION ANSWERING DATASETS

For the pretraining phase, the Discrete Reasoning Over the Content of Paragraphs (DROP) dataset [14] was employed. The DROP dataset, designed for complex reasoning tasks including span selection, arithmetic operations, and logical inference, was used to build a foundational understanding of question-answer pair structures and enhance reasoning capabilities. The pretraining dataset (DROP) allows the model to develop general reasoning skills, which are then refined during the fine-tuning phase on specific tasks.

Following pretraining, fine-tuning was performed on the Stanford Question Answering Dataset (SQuAD) v1.1 [15], the HotpotQA dataset [16], and the Natural Questions dataset [17]. datasets. These datasets are used to adapt the pre-trained model to specific tasks with limited labeled examples, simulating real-world scenarios where labeled data is scarce. Each fine-tuning dataset was evaluated under four few-shot settings: 16-shot, 32-shot, 64-shot, and 128-shot. In these settings, the number of training examples was deliberately limited to simulate data-scarce scenarios. These configurations were designed to evaluate the model's performance in low-resource environments and to assess its generalization capabilities across varying amounts of labeled data.

A. PRETRAINING DATASET

The Discrete Reasoning Over the Content of Paragraphs (DROP) dataset [14] is utilized during the pretraining phase of the QARR-QGF framework to support complex reasoning tasks. DROP includes question-answer pairs requiring the model to perform operations such as span selection, arithmetic operations, and logical inferences. The dataset contains 96,000 questions across 13,000 passages, providing an excellent resource for training models that need to reason across the content of paragraphs.

DROP is especially suitable for pretraining as it involves detailed reasoning tasks beyond simple question answering. Each instance in DROP consists of a question, a passage containing the answer, and the expected answer span or

operation to be performed. This makes it highly effective for teaching the model to infer answers through logical operations and span extraction.

B. FINE-TUNING DATASETS

1) STANFORD QUESTION ANSWERING DATASET (SQuAD)

The Stanford Question Answering Dataset (SQuAD) is a widely used benchmark for evaluating reading comprehension models. It requires models to extract answers directly from text passages, rather than selecting from predefined options. The dataset is constructed via crowdsourcing and is based on curated English Wikipedia articles. Passages are selected using Project Nayuki's internal PageRank algorithm, excluding short paragraphs, tables, and figures. Crowdworkers generate five question-answer pairs per paragraph, with additional answers added for evaluation robustness. Most answers are noun phrases. SQuAD contains 10,507 testing questions and supports a broad range of question types. Table 2 illustrates the diversity of questions derived from a single passage, highlighting the dataset's effectiveness in training and evaluating QA systems.

TABLE 2. Examples of different questions and answers based on one context from SQuAD.

Context: The "Big Five" game animals of Africa, that is the lion, leopard, buffalo, rhinoceros, and elephant, ... The annual animal migration occurs between June and September with millions of animals ... Two million wildebeest migrate a distance of 2,900 kilometres (1,802 mi) from the Serengeti ...	
Question	**Answer**
What are the "Big Five" animals in Kenya?	lion, leopard, buffalo, rhinoceros, and elephant
When does the annual animal migration occur?	between June and September
How far do the wildebeest migrate searching for food and water?	2,900 kilometres

2) HotpotQA

HotpotQA is a benchmark designed to evaluate multi-hop reasoning, where models must integrate information across multiple unstructured documents to answer a question. Unlike traditional single-hop or knowledge base–driven datasets, HotpotQA promotes diverse and complex question-answer pairs unconstrained by predefined schemas. The dataset is built from an English Wikipedia dump, using hyperlinks in the first paragraphs of articles to form a directed graph of entity relationships. Questions are constructed by leveraging bridge entities or comparing entities within the same category. HotpotQA includes 5,901 testing questions. Table 3 presents examples illustrating both bridge-based and comparative reasoning.

3) NATURAL QUESTIONS (NQ)

The Natural Questions (NQ) dataset captures real-world queries sourced from Google Search, making it well-suited for evaluating QA systems in practical settings. Each instance includes a question, a long answer span from a Wikipedia page, and short answer entities that directly address the query. All questions contain at least eight words and are paired with the top five Wikipedia pages returned by Google. Human annotators assess each page, selecting a long answer span and corresponding short answers for well-formed questions. The dataset contains 12,836 testing questions. Table 4 presents examples that highlight the dataset's focus on real-world, natural language questions.

TABLE 3. Examples of multi-hop reasoning from HotpotQA dataset.

Reasoning	Example
Inferring the bridge entity	Paragraph A: The NAB AFL Rising Star award is given annually to a standout young player in the Australian Football League. The award was won by Jesse Hogan of Melbourne who polled 49 votes, beating Carlton's Patrick Cripps who finished on 41 votes. Paragraph B: Jesse Hogan (born 12 February 1995) is a professional Australian rules footballer playing for the Melbourne Football Club in the Australian Football League (AFL). A key forward, Hogan is 1.95 m tall and weighs 100 kg… Question: What is the height of the player who won the 2015 AFL Rising Star award? Answer: 1.95 m
Comparison	Paragraph A: Marco Da Silva (born May, 30th, 1977), is a Portuguese dancer and choreographer. He was born in Bremen, Germany… Paragraph B: Erika Girardi (born 10 July 1971), known professionally as Erika Jayne, is an American singer, actress and television personality… Question: Who was born first, Erika Jayne or Marco Da Silva? Answer: Erika Jayne

TABLE 4. Examples from natural questions dataset.

Context	Question	Answer
"You 'll Never Walk Alone" is a show tune from the 1945 Rodgers and Hammerstein musical Carousel. In the second act of the musical, Nettie Fowler, the cousin …	When was You 'll never walk alone first released?	1945
After a player has been designated for assignment, the other 30 NHL teams can put in a claim or waive their claim for that player. The claims process starts at noon Eastern Time and ends 24 hours later …	How long do NHL players stay on waivers?	24 hours

V. MODEL PRETRAINING AND FINE-TUNING PROCESS

In this study, three language models are employed to evaluate the performance of the proposed QARR-QGF framework in few-shot question answering. The selected models are Longformer-Encoder-Decoder (LED) [18], BART [19], and Text-to-Text Transfer Transformer (T5). The language models are initially pretrained on the QARR-augmented DROP dataset to enhance their foundational understanding of question-answering tasks. Following this pretraining phase, the models undergo fine-tuning on a selection of few-shot training samples derived from prominent question-answering datasets, namely SQuAD, HotpotQA, and NQ. Prior to fine-tuning, the few-shot training samples are augmented using the QGF module, ensuring a richer training experience.

During the pretraining and fine-tuning processes, the language models receive the input for the training samples structured as a triplet, consisting of the question, a masked answer, and the relevant context. This input can be expressed as follows:

$$\mathbf{x}_{\text{train}} = [\text{Question: } \mathbf{q}?, \text{Answer: } \mathcal{M}, \text{Context: } \mathbf{c}]$$

Here, $\mathbf{q}$ represents the original question, $\mathcal{M}$ denotes the mask token corresponding to the missing answer, and $\mathbf{c}$ signifies the contextual information pertinent to the question.

The model's primary objective is to generate both the question and the answer in alignment with the designated target template established during the pretraining and fine-tuning phases. The output can be articulated as:

$$\mathbf{y}_{\text{train}} = [\text{Question: } \mathbf{q}?, \text{Answer: } \mathbf{a}]$$

where $\mathbf{a}$ is the generated answer.

The overarching objective during both the pretraining and fine-tuning stages is to minimize the discrepancy between the model's predicted sequence and the target sequence, thereby enabling the model to accurately infer the correct answer or fill in missing information. This training task can be formalized as the minimization of the cross-entropy loss between the predicted sequence $\hat{\mathbf{y}}$ and the target sequence $\mathbf{y}$, defined as follows:

$$\mathcal{L} = -\sum_{i=1}^{T} \mathbf{y}_i \log p(\hat{\mathbf{y}}_i | \mathbf{x}, \theta)$$

In this equation, T denotes the length of the sequence, $\mathbf{y}_i$ represents the i-th token in the target sequence, and $p(\hat{\mathbf{y}}_i|\mathbf{x}, \theta)$ indicates the model's predicted probability for the token $\hat{\mathbf{y}}_i$ given the input sequence $\mathbf{x}$, with θ symbolizing the model parameters. The predicted sequence $\hat{\mathbf{y}}$ consists of the tokens generated by the model during inference, reflecting the output that the model aims to approximate to the target sequence $\mathbf{y}$.

After the fine-tuning process, the performance of the models is evaluated on the test samples from the aforementioned question-answering datasets. The input for the test samples is structured similarly to the training inputs, represented as follows:

$$\mathbf{x}_{\text{test}} = [\text{Question: } \mathbf{q}?, \text{Answer: } \mathcal{M}, \text{Context: } \mathbf{c}]$$

where $\mathbf{x}_{\text{test}}$ contains the test question, a mask token for the missing answer, and the relevant context. The predicted output for the test samples can be defined as:

$$\mathbf{y}_{\text{predicted}} = [\text{Question: } \mathbf{q}?, \text{Answer: } \mathbf{a}_{\text{predicted}}]$$

where $\mathbf{y}_{\text{predicted}}$ represents the predicted output, including the question and the predicted answer $\mathbf{a}_{\text{predicted}}$.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

This research evaluates three generative models: LED, BART, and T5, across three diverse datasets: SQuAD, HotpotQA, and NQ. Each model is fine-tuned on each dataset using four different sample sizes: 16-shot, 32-shot, 64-shot, and 128-shot.

A. HYPERPARAMETER TUNING

Hyperparameter tuning plays a critical role in optimizing model performance by identifying the most effective configurations. In this study, two key hyperparameters are investigated: the removal proportion p and the filtering threshold τ. The removal proportion p specifies the fraction of words removed from sentences in the QARR module, while the filtering threshold τ determines the confidence level for retaining generated sentences in the QGF module.

To assess the impact of these hyperparameters, experiments were conducted with p values of 0.1, 0.25, and 0.5, and τ values of 0.98 and 0.99, resulting in six hyperparameter settings. These experiments aim to identify the optimal configuration for improving performance on benchmark datasets.

1) QARR-QGF-LED

Table 5 presents the hyperparameter tuning results for the QARR-QGF-LED model under various settings. The removal proportion p is found to be optimal at 0.1 in most configurations, indicating that the QARR-QGF-LED model benefits from retaining more complete sentences during the QARR phase.

TABLE 5. Hyperparameter tuning results for word removal proportion p and filtering threshold τ for the QARR-QGF-LED model.

Few-shot Setting	p	τ	SQuAD	HotpotQA	NQ
16-shot	0.1	0.98	**75.9**	58.7	54.6
	0.1	0.99	74.9	**59.5**	**55.4**
	0.25	0.98	74.9	57.8	54.5
	0.25	0.99	74.3	58.1	54.7
	0.5	0.98	74.9	57.0	54.3
	0.5	0.99	74.2	57.1	54.9
32-shot	0.1	0.98	**77.1**	60.0	**55.7**
	0.1	0.99	76.9	**60.3**	55.7
	0.25	0.98	76.5	59.4	55.5
	0.25	0.99	76.4	59.8	55.6
	0.5	0.98	76.2	58.5	55.6
	0.5	0.99	76.3	58.9	54.9
64-shot	0.1	0.98	77.7	61.4	**57.8**
	0.1	0.99	78.1	**61.6**	57.2
	0.25	0.98	77.4	60.8	57.3
	0.25	0.99	78.0	60.9	56.7
	0.5	0.98	77.3	59.7	57.1
	0.5	0.99	**78.1**	60.7	57.0
128-shot	0.1	0.98	79.4	63.1	**60.4**
	0.1	0.99	**80.1**	**63.7**	59.7
	0.25	0.98	79.5	62.8	60.1
	0.25	0.99	80.1	63.6	60.1
	0.5	0.98	79.1	62.5	59.5
	0.5	0.99	80.0	63.1	59.0

In the 16-shot setting, the model achieves an F1 score of 75.9% on the SQuAD dataset with τ set to 0.98, and scores of 59.5% and 55.4% on the HotpotQA and NQ datasets, respectively, with τ set to 0.99. In the 32-shot setting, the F1

scores improve to 77.1% on the SQuAD dataset and 55.7% on the NQ dataset, both with τ set to 0.98, while the model achieves an F1 score of 60.3% on the HotpotQA dataset with τ set to 0.99.

In the 64-shot setting, the model achieves an F1 score of 78.1% on the SQuAD dataset with p set to 0.5 and τ set to 0.99. The model also achieves scores of 61.6% on the HotpotQA dataset with τ set to 0.99 and 57.8% on the NQ dataset with τ set to 0.98. In the 128-shot setting, the model reaches its highest scores, achieving 80.1% on the SQuAD dataset and 63.7% on the HotpotQA dataset, both with τ set to 0.99, and 60.4% on the NQ dataset with τ set to 0.98.

2) QARR-QGF-BART

Table 6 presents the hyperparameter tuning results for the word removal proportion p and the filtering threshold τ on the BART models across various shot settings. The results indicate that the optimal hyperparameters for the model vary depending on the dataset and the number of training samples.

TABLE 6. Hyperparameter tuning results for word removal proportion p and filtering threshold τ for the QARR-QGF-BART model.

Few-shot Setting	p	τ	SQuAD	HotpotQA	NQ
16-shot	0.1	0.98	75.5	57.8	54.9
	0.1	0.99	**76.1**	**58.4**	54.7
	0.25	0.98	75.6	57.3	**55.5**
	0.25	0.99	75.3	58.4	55.3
	0.5	0.98	75.9	56.9	55.4
	0.5	0.99	75.7	57.1	55.0
32-shot	0.1	0.98	77.1	59.3	55.8
	0.1	0.99	77.4	**59.5**	55.9
	0.25	0.98	77.1	59.0	**56.4**
	0.25	0.99	**77.6**	59.5	56.3
	0.5	0.98	76.9	58.5	56.2
	0.5	0.99	77.0	58.9	56.1
64-shot	0.1	0.98	78.0	60.9	57.5
	0.1	0.99	77.7	61.6	57.0
	0.25	0.98	77.7	61.0	**58.2**
	0.25	0.99	78.4	**61.7**	57.3
	0.5	0.98	78.1	60.5	57.9
	0.5	0.99	**78.5**	61.1	57.3
128-shot	0.1	0.98	79.7	62.5	60.7
	0.1	0.99	80.3	63.4	59.7
	0.25	0.98	79.9	62.6	60.9
	0.25	0.99	**80.8**	**63.6**	**61.0**
	0.5	0.98	80.0	62.3	60.4
	0.5	0.99	80.6	63.0	60.6

In the 16-shot setting, the optimal values are $p = 0.1$ and $\tau = 0.99$, achieving F1 scores of 76.1% and 58.4% on the SQuAD and HotpotQA datasets, respectively. For the NQ dataset, the model achieves an F1 score of 55.5%, with the optimal values being $p = 0.25$ and $\tau = 0.98$. In the 32-shot setting, the QARR-QGF-BART model demonstrates improved performance, achieving an F1 score of 77.6% on the SQuAD dataset with $p = 0.25$ and $\tau = 0.99$. The model achieves an F1 score of 59.5% on the HotpotQA dataset with $p = 0.1$ and $\tau = 0.99$, while on the NQ dataset, it achieves an F1 score of 56.4% with $p = 0.25$ and $\tau = 0.98$.

In the 64-shot setting, the model achieves an F1 score of 78.5% on the SQuAD dataset with $p = 0.5$ and $\tau = 0.99$. For the HotpotQA dataset, the model achieves 61.7% with $p = 0.25$ and $\tau = 0.99$, while on the NQ dataset, it achieves 58.2% with $p = 0.25$ and $\tau = 0.98$. In the 128-shot setting, the optimal values are $p = 0.25$ and $\tau = 0.99$, achieving F1 scores of 80.8%, 63.6%, and 61.0% on the SQuAD, HotpotQA, and NQ datasets, respectively.

3) QARR-QGF-T5

Table 7 presents the hyperparameter tuning results for the word removal proportion p and the filtering threshold τ on the T5 models. Based on the results, the best value for p in the T5 model is consistently 0.25 across all experimental settings. This moderate proportion effectively removes words while preserving the semantic meaning of the original sentence.

TABLE 7. Hyperparameter tuning results for word removal proportion p and filtering threshold τ for the QARR-QGF-T5 model.

Few-shot Setting	p	τ	SQuAD	HotpotQA	NQ
16-shot	0.1	0.98	76.1	57.2	54.3
	0.1	0.99	75.8	57.1	54.0
	0.25	0.98	82.1	**59.9**	58.4
	0.25	0.99	**82.3**	59.8	**59.0**
	0.5	0.98	81.4	59.1	57.6
	0.5	0.99	81.2	59.1	58.1
32-shot	0.1	0.98	77.4	59.2	54.6
	0.1	0.99	77.8	59.4	55.3
	0.25	0.98	82.5	60.3	60.4
	0.25	0.99	**83.0**	**60.8**	**61.1**
	0.5	0.98	81.5	60.2	59.7
	0.5	0.99	82.2	59.9	59.0
64-shot	0.1	0.98	78.1	61.4	59.0
	0.1	0.99	79.0	61.6	58.7
	0.25	0.98	82.4	62.1	62.2
	0.25	0.99	**83.4**	**62.5**	**62.3**
	0.5	0.98	81.8	61.6	61.1
	0.5	0.99	82.6	62.3	61.6
128-shot	0.1	0.98	80.5	62.3	61.5
	0.1	0.99	81.4	63.6	60.4
	0.25	0.98	84.2	63.4	**64.7**
	0.25	0.99	**84.7**	**64.1**	64.2
	0.5	0.98	83.8	62.9	64.2
	0.5	0.99	84.5	63.2	63.0

On the HotpotQA dataset in the 16-shot setting, the optimal τ is 0.98, achieving an F1 score of 59.9%. The QARR-QGF-T5 model achieves an F1 score of 82.3% on the SQuAD dataset and 59.0% on the Natural Questions (NQ) dataset. In the 32-shot setting, the optimal τ is 0.99, achieving F1 scores of 83.0% on the SQuAD dataset, 60.8% on the HotpotQA dataset, and 61.1% on the NQ dataset.

In the 64-shot setting, similar to the previous configurations, the optimal τ remains 0.99, achieving F1 scores

of 83.4% on the SQuAD dataset, 62.5% on the HotpotQA dataset, and 62.3% on the NQ dataset. In the 128-shot setting, the QARR-QGF-T5 model performs better with τ set to 0.98, achieving an F1 score of 64.7% on the NQ dataset, while on the SQuAD and HotpotQA datasets, the model achieves F1 scores of 84.7% and 64.1%, respectively. The higher filtering threshold in these settings effectively removes low-quality generated questions, maintaining the overall quality of the question-answer pairs in the training set.

B. ABLATION STUDY

An ablation study was conducted to evaluate the impact of the QARR module and QGF module on the LED, BART, and T5 models across four different shot settings on the SQuAD, HotpotQA, and NQ datasets. The results are summarized in Table 8. The base models (LED, BART, and T5) were pretrained on the original DROP dataset and fine-tuned on the respective question answering datasets. The QARR-augmented models (QARR-LED, QARR-BART, and QARR-T5) were pretrained on the QARR-augmented DROP dataset and fine-tuned on the original question answering datasets. Finally, the QARR-QGF models (QARR-QGF-LED, QARR-QGF-BART, and QARR-QGF-T5) were pretrained on the QARR-augmented DROP dataset and fine-tuned on the QGF-augmented question answering datasets. The results demonstrate that incorporating the QARR-QGF framework consistently yields incremental improvements across datasets and architectures.

Among the evaluated models, the QARR-QGF-T5 achieves the best performance in all shot settings. For instance, in the 16-shot setting on the SQuAD dataset, integrating the QARR module into the base T5 model improves the F1 score from 77.5% to 81.7%. Adding the QGF module further enhances the F1 score to 82.3%. Similar trends are observed for the HotpotQA and NQ datasets, where the QARR-QGF-T5 model achieves F1 scores of 59.9% and 59.0%, respectively. In the 32-shot setting, the T5 model exhibits further improvements with the integration of both modules. The base T5 model initially scores 79.5% on the SQuAD dataset. Incorporating the QARR module increases the F1 score to 81.7%, while the addition of the QGF module further improves it to 83.0%. Similar patterns are observed in the HotpotQA and NQ datasets.

The performance of the models steadily improves as the number of training samples increases from 16-shot to 128-shot settings, reflecting their ability to generalize better with additional data. For example, on the SQuAD dataset, the QARR-QGF-T5 model achieves an F1 score of 82.3% in the 16-shot setting, which rises to 83.0% in the 32-shot setting, 83.4% in the 64-shot setting, and 84.7% in the 128-shot setting. These results suggest that the proposed QARR and QGF modules contribute positively to improving model robustness in few-shot learning scenarios by enhancing data diversity and training stability even with limited supervision.

TABLE 8. Comparative results of base models, QARR models, and QARR-QGF models in terms of F1 scores across few-shot settings.

Shots	Model	SQuAD	HotpotQA	NQ
16-shot	LED	74.7	57.3	52.3
	QARR-LED	75.8	59.3	53.3
	QARR-QGF-LED	75.9	59.5	55.4
	BART	75.0	56.9	54.1
	QARR-BART	75.2	58.3	54.9
	QARR-QGF-BART	76.1	58.4	55.5
	T5	77.5	55.7	56.7
	QARR-T5	81.7	59.0	56.7
	QARR-QGF-T5	**82.3**	**59.9**	**59.0**
32-shot	LED	75.2	57.7	53.4
	QARR-LED	76.6	60.0	54.6
	QARR-QGF-LED	77.1	60.3	55.7
	BART	75.5	56.9	55.0
	QARR-BART	76.3	59.2	56.4
	QARR-QGF-BART	77.6	59.5	56.4
	T5	79.5	58.5	57.0
	QARR-T5	81.7	60.2	57.1
	QARR-QGF-T5	**83.0**	**60.8**	**61.1**
64-shot	LED	75.8	60.2	54.4
	QARR-LED	77.1	61.0	55.2
	QARR-QGF-LED	78.1	61.6	57.8
	BART	75.1	58.7	55.6
	QARR-BART	76.9	61.1	57.8
	QARR-QGF-BART	78.5	61.7	58.2
	T5	80.5	59.9	59.3
	QARR-T5	82.7	61.5	60.1
	QARR-QGF-T5	**83.4**	**62.5**	**62.3**
128-shot	LED	79.1	61.9	56.4
	QARR-LED	79.1	63.0	57.8
	QARR-QGF-LED	80.8	63.7	60.4
	BART	79.4	62.2	59.3
	QARR-BART	80.5	63.3	61.0
	QARR-QGF-BART	80.8	63.6	61.0
	T5	81.1	62.3	61.1
	QARR-T5	84.5	63.6	63.0
	QARR-QGF-T5	**84.7**	**64.1**	**64.7**

C. QUANTITATIVE EVALUATION

This section presents an in-depth analysis of the errors encountered in the QARR-QGF model's performance. Table 9 categorizes the errors observed during an experiment conducted with the QARR-QGF-T5 model in a 128-shot setting. Answers were deemed correct only if they exactly matched the gold standard, resulting in 74.52% of the predictions being classified as correct. This reflects the model's strong overall performance but also highlights areas for further improvement. Among the 25.48% incorrect answers, 8.25% were categorized as over-informative, where the model provided more information than necessary. Conversely, 4.19% of the errors were incomplete answers, omitting critical details required for accurate answers. Numeric errors, though rare at 0.02%, typically involved formatting discrepancies, such as providing a numeral instead

of its written form. The largest category of incorrect answers, unrelated answers, accounted for 13.02% of the total errors, where the model generated outputs completely irrelevant to the question. To further illustrate the nature of these errors, Table 10 provides representative examples from each category. These examples demonstrate the specific challenges faced by the QARR-QGF-T5 model, offering insights into areas for potential improvement.

TABLE 9. Error analysis of QARR-QGF-T5 in the 128-shot setting.

Category	Percentage (%)
Correct Answer	**74.52**
Incorrect Answer	**25.48**
• Over-Informative	8.25
• Incomplete Answer	4.19
• Numeric Problem	0.02
• Unrelated Answer	13.02

TABLE 10. Examples of errors in the QARR-QGF-T5 model with actual and predicted answers.

Error Category	Question	Actual Answer	Predicted Answer
Over-Informative	What field type is usually chosen for Super Bowl games?	natural grass	natural grass stadiums
	How is oxygen ranked as abundant in the universe?	third	third most abundant chemical element
Incomplete Answer	When did Tesla begin researching X-ray imaging?	March 1896	1896
	What has started to change the way teachers teach in the classroom, generally?	increasing use of technology	technology
Numeric Problem	How many field goals did McManus kick in the game?	four	4
	How many customaries does Norman customary law have?	two	2
Unrelated Answer	Other than many sunny days, what characteristic is typical for the climate in Southern California?	infrequent rain	Mediterranean climate
	What were steam engines used as a source of?	power	high-pressure steam

D. QUALITATIVE EVALUATION

Table 11 presents a comparative analysis of answers generated by the baseline T5 model and the QARR-QGF-T5 model, both fine-tuned under the 128-shot setting on the SQuAD dataset. These examples highlight how the QARR-QGF framework enhances the model's ability to comprehend context, retrieve relevant information, and generate accurate answers.

In the first example, the baseline T5 model incorrectly outputs 'San Francisco Bay Area,' which refers to a broader geographical region rather than the specific venue requested in the question. This error stems from the model's inability to focus on precise details within the passage. In contrast, the QARR-QGF-T5 model correctly identifies 'Stanford Stadium' as the venue for Super Bowl XIX. The improvement demonstrates the QARR-QGF framework's capacity to refine contextual understanding, enabling the model to extract specific and accurate details.

The second example involves a more complex scenario where the baseline T5 model incorrectly generates 'Bennie Fowler' as the answer. The error arises due to confusion between two closely related events: Anderson scoring a 2-yard touchdown and Manning completing a pass to Bennie Fowler for a 2-point conversion. These events occur sequentially within the passage and are linked to the same scoring sequence, causing the baseline model to mistake one for the other. The QARR-QGF-T5 model, however, successfully resolves this ambiguity and generates 'Anderson' as the correct answer, illustrating its enhanced ability to distinguish between related events and maintain contextual clarity.

In the third example, the question asks about the type of forested areas on the highest terrace. The baseline T5 model incorrectly produces 'peat swamps' as the answer. This mistake arises because the model fails to focus on the key term 'forested areas', instead concentrating solely on descriptions of the highest terrace. In contrast, the QARR-QGF-T5 model generates the correct answer, 'pine', demonstrating its ability to align the generated response with the specific semantic requirements of the question.

These case studies underscore the effectiveness of the QARR-QGF framework in addressing challenges such as ambiguity, contextual misinterpretation, and the need for precise information extraction. By augmenting both the pretraining and fine-tuning phases with high-quality, diverse, and contextually relevant data, the QARR-QGF framework significantly improves the model's robustness and accuracy in few-shot question answering tasks.

E. COMPARATIVE RESULTS WITH EXISTING WORK

The proposed QARR-QGF framework demonstrates superior performance in few-shot question answering tasks across the SQuAD, HotpotQA, and NQ datasets compared to existing methods. The empirical results, summarized in Table 12, validate the effectiveness of the QARR-QGF framework. To ensure statistical robustness of the reported results, each few-shot experiment was conducted over five independent runs with different random seeds, reflecting variability in few-shot sample selection. The reported performance values represent the mean and standard deviation across these runs, following common practices adopted in existing few-shot learning studies. This approach accounts for the inherent stochasticity of few-shot training and allows a more reliable comparison of model performance.

In the 16-shot setting on the SQuAD dataset, the QARR-QGF-T5 model achieves an F1 score of 82.3%, compared to 46.5% achieved by PMR [2], 55.5% by FewShotQA [1], 56.1% by Resilient Model [12], 57.8% by GOTTA [6], and 65.9% by TPKE-QA [4]. Similarly, on the HotpotQA

TABLE 11. Examples with and without the QARR-QGF framework, showing the context, question, and answers from the baseline T5 model and the QARR-QGF-T5 model.

Context	Question	Gold Answer	Answer by Baseline T5	Answer by QARR-QGF-T5
The league announced on October 16, 2012, that the two finalists were Sun Life Stadium and Levi's Stadium. The South Florida/Miami area has previously hosted the event 10 times (tied for most with New Orleans), with the most recent one being Super Bowl XLIV in 2010. The San Francisco Bay Area last hosted in 1985 (Super Bowl XIX), held at Stanford Stadium in Stanford, California, won by the home team 49ers. The Miami bid depended on whether the stadium underwent renovations. However, on May 3, 2013, the Florida legislature refused to approve the funding plan to pay for the renovations, dealing a significant blow to Miami's chances.	In what venue did Super Bowl XIX take place?	Stanford Stadium	San Francisco Bay Area	Stanford Stadium
With 4:51 left in regulation, Carolina got the ball on their own 24-yard line with a chance to mount a game-winning drive, and soon faced 3rd-and-9. On the next play, Miller stripped the ball away from Newton, and after several players dove for it, it took a long bounce backwards and was recovered by Ward, who returned it five yards to the Panthers 4-yard line. Although several players dove into the pile to attempt to recover it, Newton did not and his lack of aggression later earned him heavy criticism. Meanwhile, Denver's offense was kept out of the end zone for three plays, but a holding penalty on cornerback Josh Norman gave the Broncos a new set of downs. Then Anderson scored on a 2-yard touchdown run and Manning completed a pass to Bennie Fowler for a 2-point conversion, giving Denver a 24-10 lead with 3:08 left and essentially putting the game away. Carolina had two more drives, but failed to get a first down on each one.	What is the last name of the player who scored a 2-yard touchdown in the fourth quarter?	Anderson	Bennie Fowler	Anderson
The plain moraine plateau has only a few natural and artificial ponds and also groups of clay pits. The pattern of the Vistula terraces is asymmetrical. The left side consists mainly of two levels: the highest one contains former flooded terraces and the lowest one the flood plain terrace. The contemporary flooded terrace still has visible valleys and ground depressions with water systems coming from the Vistula old riverbed. They consist of still quite natural streams and lakes as well as the pattern of drainage ditches. The right side of Warsaw has a different pattern of geomorphological forms. There are several levels of the plain Vistula terraces (flooded as well as former flooded ones) and only a small part and not-so-visible moraine escarpment. Aeolian sand with a number of dunes parted by peat swamps or small ponds covers the highest terrace. These are mainly forested areas (pine forest).	What type of forested areas can be found on the highest terrace?	pine	peat swamps	pine

and NQ datasets, the QARR-QGF-T5 model achieves F1 scores of 59.9% and 59.0%, respectively, outperforming all baseline methods. Consistent improvements in performance are observed across all shot settings. For instance, as the number of training samples increases from 16-shot to 128-shot, the F1 score on the SQuAD dataset rises from 82.3% to 84.7%, demonstrating the model's ability to leverage additional training data effectively. These results emphasize the framework's capability to address the challenges posed by limited training data effectively. The improvements can be attributed to the distinct contributions of the QARR and QGF modules, both of which play complementary roles in data augmentation.

The QARR module focuses on augmenting the pretraining dataset by creating diverse and semantically coherent question-answer pairs. This is achieved through two key processes: question-answer replacement and word removal. The question-answer replacement process modifies existing questions by substituting specific elements with corresponding answers, thereby generating alternative phrasings of the same semantic intent. This approach enhances lexical and syntactic diversity, enabling the model to recognize varied linguistic patterns. Conversely, the word removal process selectively removes parts of the question while treating the removed segment as the answer. This strategy introduces naturally occurring incomplete or elliptical questions, improving the model's robustness to variations in question formulations. By enriching the pretraining dataset with such variations, the QARR module equips the model with a generalized understanding of question-answer relationships.

The QGF module addresses the limitations of few-shot learning during fine-tuning by further augmenting the training dataset with high-quality paraphrased questions. This module operates in two stages: question generation and question filtering. In the question generation stage, a sequence-to-sequence model, such as BART, generates paraphrased versions of existing questions. These paraphrases introduce linguistic variability while maintaining semantic equivalence, thereby enhancing the diversity of

TABLE 12. Comparative results with existing works.

Few-Shot	Method	SQuAD	HotpotQA	NQ
16-shot	FewShotQA [1]	55.5±2.0	45.1±1.8	45.1±2.3
	PMR [2]	46.5±4.0	32.9±2.9	32.6±3.0
	TPKE-QA [4]	65.9±3.4	49.3±2.5	40.3±4.1
	GOTTA [6]	57.8±2.6	45.9±1.7	47.1±1.1
	Resilient Model [12]	56.1±3.6	45.5±0.9	46.9±1.2
	QARR-QGF-LED	75.9±0.9	59.5±0.4	55.4±0.8
	QARR-QGF-BART	76.1±0.9	58.4±1.5	55.5±1.4
	QARR-QGF-T5	**82.3±0.7**	**59.9±1.1**	**59.0±1.1**
32-shot	FewShotQA [1]	56.8±2.1	47.9±1.4	50.1±1.1
	PMR [2]	61.0±2.9	43.5±0.8	41.6±3.6
	TPKE-QA [4]	70.5±2.7	55.1±2.1	46.0±1.6
	GOTTA [6]	62.7±1.8	49.6±1.3	49.6±1.3
	Resilient Model [12]	63.3±1.6	49.1±1.5	50.4±2.1
	QARR-QGF-LED	77.1±1.3	60.3±1.0	55.7±2.4
	QARR-QGF-BART	77.6±0.7	59.5±1.4	56.4±2.2
	QARR-QGF-T5	**83.0±0.3**	**60.8±1.3**	**61.1±0.7**
64-shot	FewShotQA [1]	61.5±2.3	51.2±1.0	53.0±0.5
	PMR [2]	-	-	-
	TPKE-QA [4]	74.9±0.5	59.3±1.7	50.4±2.1
	GOTTA [6]	67.7±0.9	52.0±0.8	51.5±1.3
	Resilient Model [12]	68.2±0.8	52.3±1.3	52.4±0.8
	QARR-QGF-LED	78.1±0.9	61.6±1.2	57.8±1.8
	QARR-QGF-BART	78.5±1.2	61.7±1.2	58.2±1.8
	QARR-QGF-T5	**83.4±0.8**	**62.5±1.1**	**62.3±1.1**
128-shot	FewShotQA [1]	68.0±0.3	54.8±0.8	53.9±0.9
	PMR [2]	73.1±0.9	56.4±1.5	51.9±1.7
	TPKE-QA [4]	77.1±0.5	62.8±1.2	55.9±1.0
	GOTTA [6]	71.3±1.3	56.3±1.4	54.2±0.7
	Resilient Model [12]	73.1±0.6	57.3±0.5	54.6±0.1
	QARR-QGF-LED	80.8±0.7	63.7±1.1	60.4±0.3
	QARR-QGF-BART	80.8±0.6	63.6±1.0	61.0±0.4
	QARR-QGF-T5	**84.7±0.3**	**64.1±0.7**	**64.7±0.6**

the fine-tuning dataset. This variability reduces the risk of overfitting to the small original dataset and reinforces the model's ability to generalize to unseen questions. To ensure the quality of the augmented questions, the question filtering stage applies a semantic similarity-based mechanism. Both the original and generated questions are encoded into a shared vector space, and their cosine similarity is computed. A threshold is then applied to retain only paraphrases that closely align with the semantic intent of the original question. By filtering out low-quality or noisy samples, the QGF module maintains the integrity of the augmented dataset. This ensures that the model is trained on high-quality, contextually relevant question-answer pairs, which is crucial for effective few-shot learning.

The synergy between the QARR and QGF modules is a key factor in the framework's success. The QARR module enriches the pretraining dataset, providing a strong foundation for the model, while the QGF module enhances the fine-tuning dataset, enabling the model to adapt effectively to specific tasks. Together, these modules address the challenges of data scarcity in few-shot learning by systematically increasing the quantity, diversity, and quality of the training data. This two-stage augmentation pipeline ensures that the model is exposed to a wide range of linguistic structures and semantic variations, leading to superior generalization capabilities.

VII. CONCLUSION

This paper presents the QARR-QGF framework, a dual-module data augmentation strategy for addressing the challenges of few-shot question answering. The framework systematically integrates the QARR module and the QGF module to enhance the diversity, quality, and robustness of training data during the pretraining and fine-tuning phases, respectively. By addressing the limitations of existing methods, such as overfitting, semantic inconsistency, and insufficient data augmentation, the QARR-QGF framework demonstrates its efficacy in improving model generalization and adaptability in low-resource scenarios.

Comprehensive evaluations on three widely used few-shot question answering datasets, namely SQuAD, HotpotQA, and Natural Questions, highlight the superior performance of the QARR-QGF framework compared to state-of-the-art approaches. The proposed framework consistently achieves higher F1 scores across various few-shot settings (16-shot, 32-shot, 64-shot, and 128-shot), emphasizing its capability to effectively leverage limited training data. The results also underscore the complementary roles of the QARR and QGF modules, with the former enriching the pretraining phase through systematic data diversification and the latter augmenting fine-tuning with high-quality, semantically filtered paraphrased questions.

Future work could explore extending the QARR-QGF framework to other natural language processing tasks, such as summarization, translation, and dialogue systems, where similar low-resource challenges are prevalent. Furthermore, investigating adaptive mechanisms for optimizing the parameters of the QARR and QGF modules could enhance the framework's efficiency and scalability. Another potential direction is to assess the framework's robustness under domain shift conditions, including out-of-distribution datasets and multilingual question answering, to further validate its generalization capability beyond the English-language datasets used in this study. Lastly, integrating external knowledge sources or hybrid retrieval-augmented strategies may further improve the framework's ability to handle complex reasoning tasks and domain-specific questions.

In conclusion, the QARR-QGF framework provides a practical and effective augmentation strategy for few-shot question answering, offering a robust and generalizable solution to the challenges of limited data scenarios. Its modular design, consistent empirical improvements, and extensibility make it a valuable contribution to the field of natural language processing, with potential implications for a wide range of applications in low-resource scenarios.

REFERENCES

[1] R. Chada and P. Natarajan, "FewshotQA: A simple framework for few-shot learning of question answering tasks using pre-trained text-to-text models," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2021, pp. 6081–6090.
[2] W. Xu, X. Li, W. Zhang, M. Zhou, W. Lam, L. Si, and L. Bing, "From cloze to comprehension: Retrofitting pre-trained masked language models to pre-trained machine reader," *Adv. Neural Inf. Process. Syst.*, vol. 36, 2024.
[3] R. Li, Q. Xiao, J. Yang, L. Zhang, and Y. Chen, "MRC-PASCL: A few-shot machine reading comprehension approach via post-training and answer span-oriented contrastive learning," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 32, pp. 4838–4849, 2024.
[4] Q. Xiao, R. Li, J. Yang, Y. Chen, S. Jiang, and D. Wang, "TPKE-QA: A gapless few-shot extractive question answering approach via task-aware post-training and knowledge enhancement," *Expert Syst. Appl.*, vol. 254, Nov. 2024, Art. no. 124475.
[5] O. Ram, Y. Kirstain, J. Berant, A. Globerson, and O. Levy, "Few-shot question answering by pretraining span selection," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics 11th Int. Joint Conf. Natural Lang. Process. (Volume 1: Long Papers)*, 2021, pp. 3066–3079.
[6] X. Chen, Y. Zhang, J. Deng, J.-Y. Jiang, and W. Wang, "Gotta: Generative few-shot question answering by prompt-based cloze data augmentation," in *Proc. SIAM Int. Conf. Data Mining (SDM)*, 2023, pp. 909–917.
[7] X. Chen, J.-Y. Jiang, W.-C. Chang, C.-J. Hsieh, H.-F. Yu, and W. Wang, "MinPrompt: Graph-based minimal prompt data augmentation for few-shot question answering," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (Long Papers)*, vol. 1, 2024, pp. 254–266.
[8] S. Haji, K. Suekane, H. Sano, and T. Takagi, "Exploratory inference chain: Exploratorily chaining multi-hop inferences with large language models for question-answering," in *Proc. IEEE 17th Int. Conf. Semantic Comput. (ICSC)*, Feb. 2023, pp. 175–182.
[9] H. Zhang, M. Cai, X. Zhang, C. Jason Zhang, R. Mao, and K. Wu, "Self-convinced prompting: Few-shot question answering with repeated introspection," 2023, *arXiv:2310.05035*.
[10] H. Luo, Y.-S. Chuang, Y. Gong, T. Zhang, Y. Kim, X. Wu, D. Fox, H. Meng, and J. Glass, "SAIL: Search-augmented instruction learning," 2023, *arXiv:2305.15225*.
[11] W. Yu, Z. Zhang, Z. Liang, M. Jiang, and A. Sabharwal, "Improving language models via plug-and-play retrieval feedback," 2023, *arXiv:2305.14002*.
[12] A. Zou, Y. Chen, R. Huang, and Y. Qin, "A resilient generative model in few-shot question answering," *Knowl.-Based Syst.*, vol. 323, Jul. 2025, Art. no. 113629.
[13] M. Lewis, Y. Liu, N. Goyal, M. Ghazvininejad, A. Mohamed, O. Levy, V. Stoyanov, and L. Zettlemoyer, "BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension," in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics*, 2020, pp. 7871–7880.
[14] D. Dua, Y. Wang, P. Dasigi, G. Stanovsky, S. Singh, and M. Gardner, "DROP: A reading comprehension benchmark requiring discrete reasoning over paragraphs," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol.*, 2019, pp. 2368–2378.
[15] P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang, "SQuAD: 100,000+ questions for machine comprehension of text," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2016, pp. 2383–2392.
[16] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. Cohen, R. Salakhutdinov, and C. D. Manning, "HotpotQA: A dataset for diverse, explainable multi-hop question answering," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2018, pp. 2369–2380.
[17] T. Kwiatkowski, J. Palomaki, O. Redfield, M. Collins, A. P. Parikh, C. Alberti, D. Epstein, I. Polosukhin, J. Devlin, K. Lee, K. Toutanova, L. Jones, M. Kelcey, M. Chang, A. M. Dai, J. Uszkoreit, Q. V. Le, and S. Petrov, "Natural questions: A benchmark for question answering research," *Trans. Assoc. Comput. Linguistics*, vol. 7, pp. 453–466, Jan. 2019.
[18] I. Beltagy, M. E. Peters, and A. Cohan, "Longformer: The long-document transformer," 2020, *arXiv:2004.05150*.
[19] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, "Exploring the limits of transfer learning with a unified text-to-text transformer," *J. Mach. Learn. Res.*, vol. 21, no. 140, pp. 1–67, 2019.



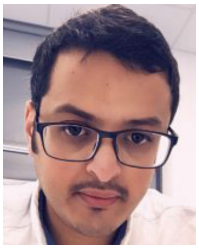
SIAO WAH TAN received the bachelor's degree (Hons.) in information technology with a specialization in artificial intelligence from Multimedia University, Malaysia, in 2022. He is currently pursuing the M.Sc. degree in information technology, focusing on natural language processing. His current research interests include question answering and few-shot learning.



CHIN POO LEE (Senior Member, IEEE) received the Master of Science and Ph.D. degrees in information technology, with a specialization in video-based abnormal behavior detection and gait recognition. She is currently an Assistant Professor with the School of Computer Science, University of Nottingham Ningbo China. Her research interests include computer vision, natural language processing, and deep learning.



KIAN MING LIM (Senior Member, IEEE) received the B.IT. degree (Hons.) in information systems engineering, the Master of Engineering Science (M.Eng.Sc.) degree, and the Ph.D. degree in IT from Multimedia University. He is currently an Associate Professor with the School of Computer Science, University of Nottingham Ningbo China. His research and teaching interests include machine learning, deep learning, computer vision, and pattern recognition.



ALI ALQAHTANI received the Ph.D. degree in computer science from Swansea University, Swansea, U.K., in 2021. He is currently an Associate Professor with the Department of Computer Science, King Khalid University, Abha, Saudi Arabia. He has published several refereed conference and journal publications. His research interests include aspects of pattern recognition, deep learning, and machine intelligence and their applications to real-world problems.